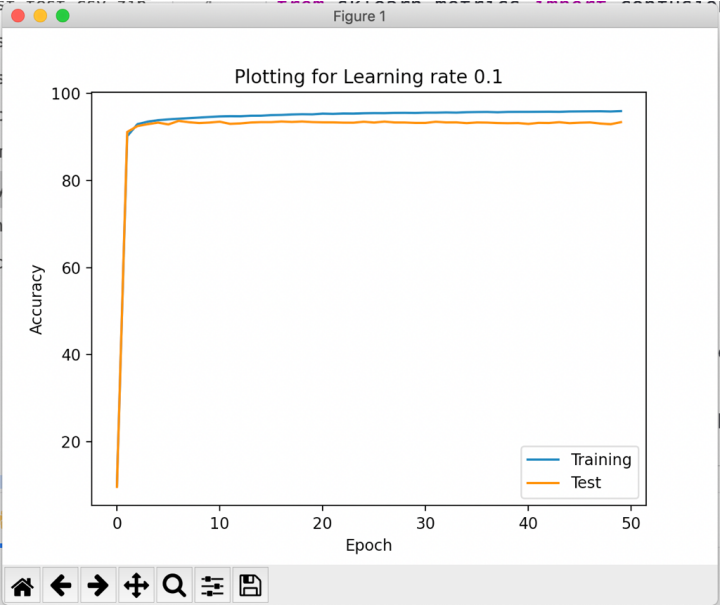


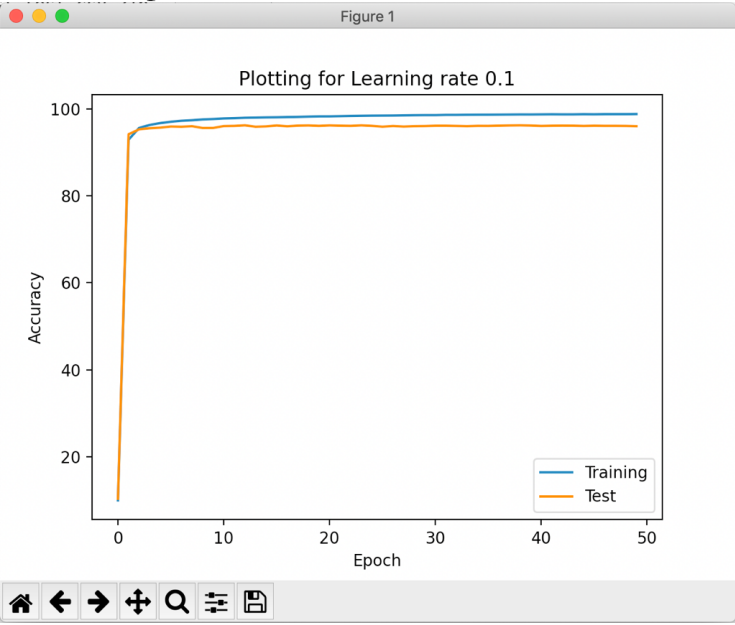
Experiment-1

1. How does the number of hidden units affect the final accuracy on the test data?
2. How does it affect the number of epochs needed for training to converge?
3. Is there evidence that any of your networks has overfit to the training data? If so, what is that evidence?

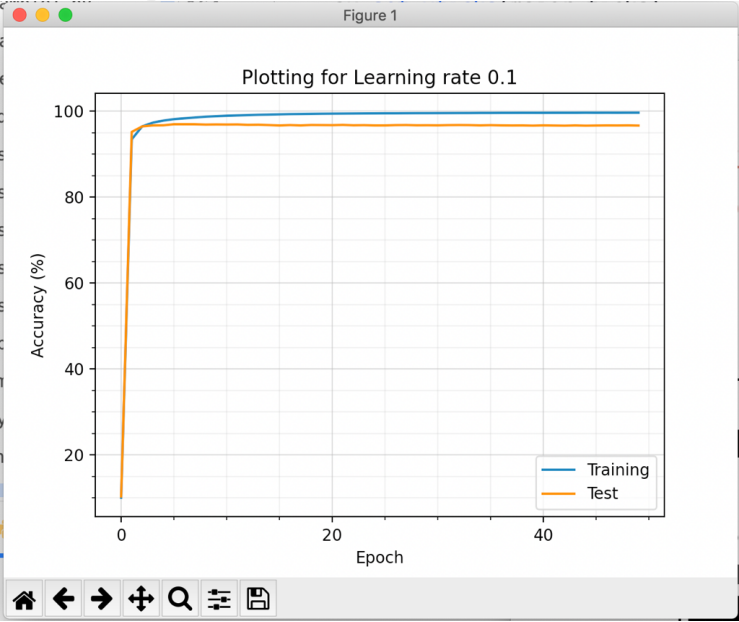
Hidden input : 20



Hidden Input : 50



Hidden Input : 100



Observation:

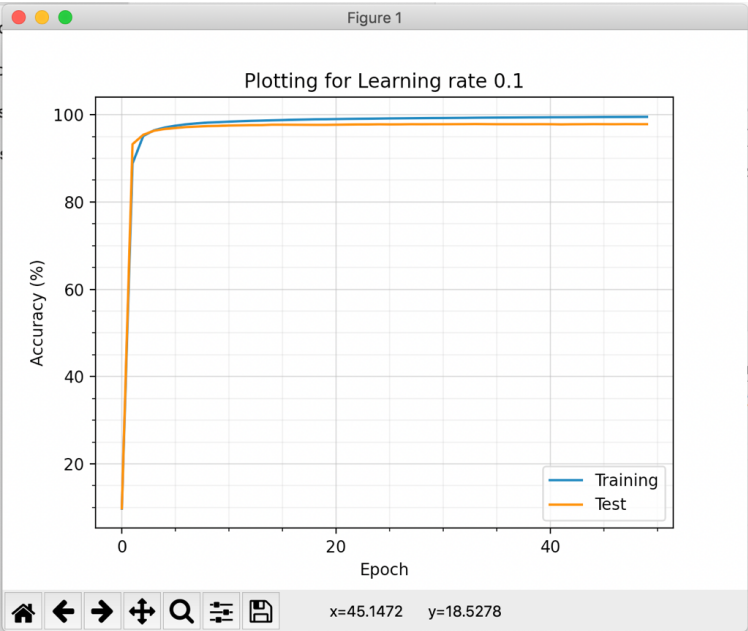
Final accuracy on test data was increased with the higher number of hidden units. Above plotting shows that accuracy is close to 100 where the number of hidden units is 100. So, higher number of hidden units will have less number of epochs to train. n hidden units 100, the training converges after 20 epochs. Networks have overfit to the training data as there is no much fluctuations between training accuracy and testing accuracy.

Experiment-2

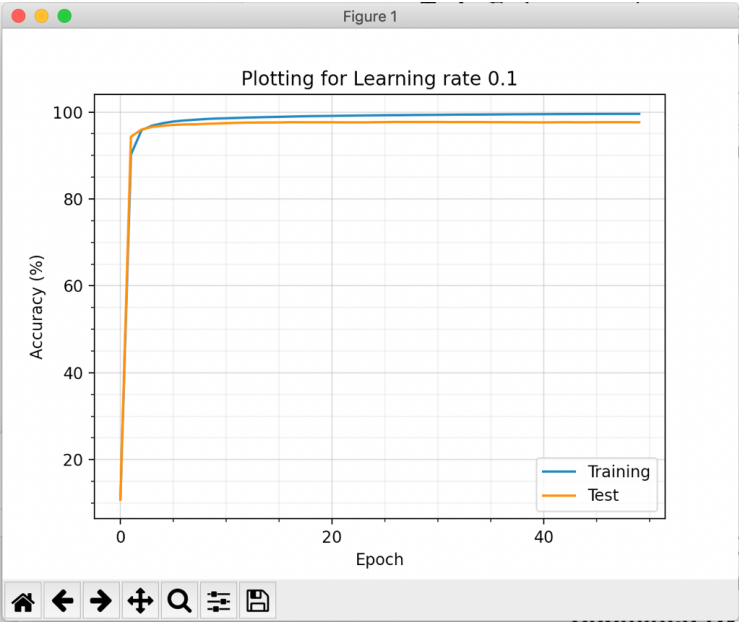
1. How does the momentum value affect the final accuracy on the test data?
2. How does it affect the number of epochs needed for training to converge?

3. Again, is there evidence that any of your networks has overfit to the training data? If so, what is that evidence?

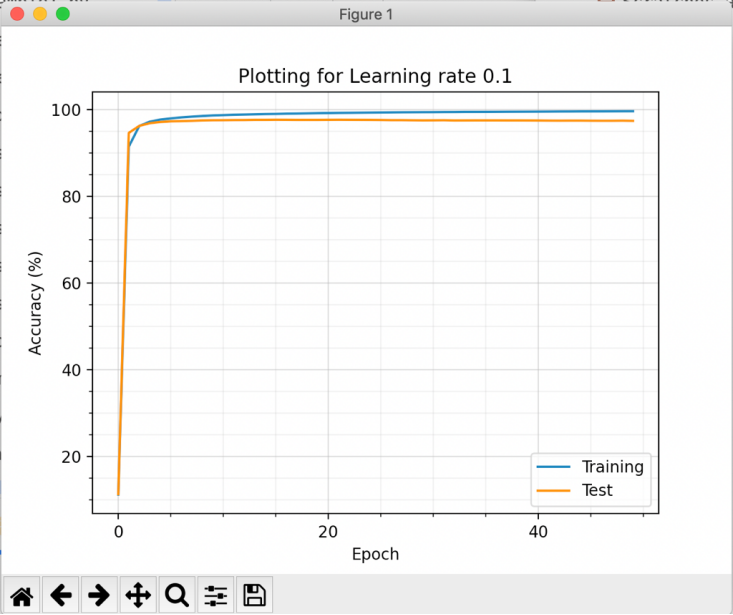
momentum : 0.0    hidden nodes : 100



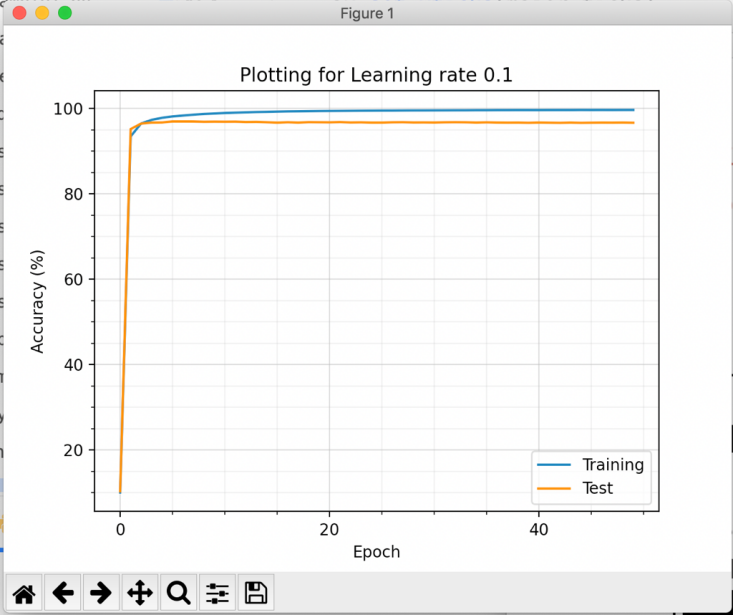
momentum : 0.25    hidden nodes : 100



momentum : 0.5    hidden nodes : 100



momentum : 0.9    hidden nodes : 100



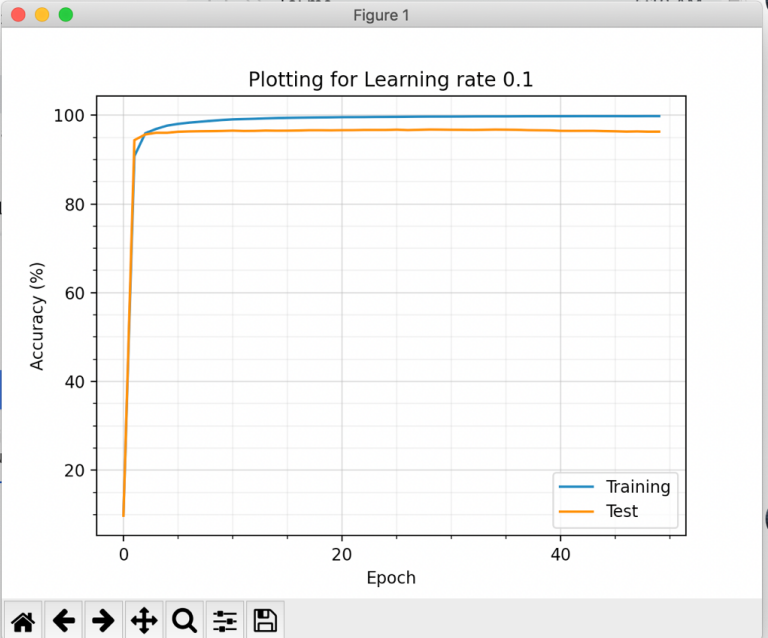
Observation:

While the smaller momentum value requires lesser number of epochs to train, the accuracy of test data gets closer to 100. In momentum 0, the training converges after 15 epochs. Networks have overfit to the training data as there is no much fluctuations between training accuracy and testing accuracy.

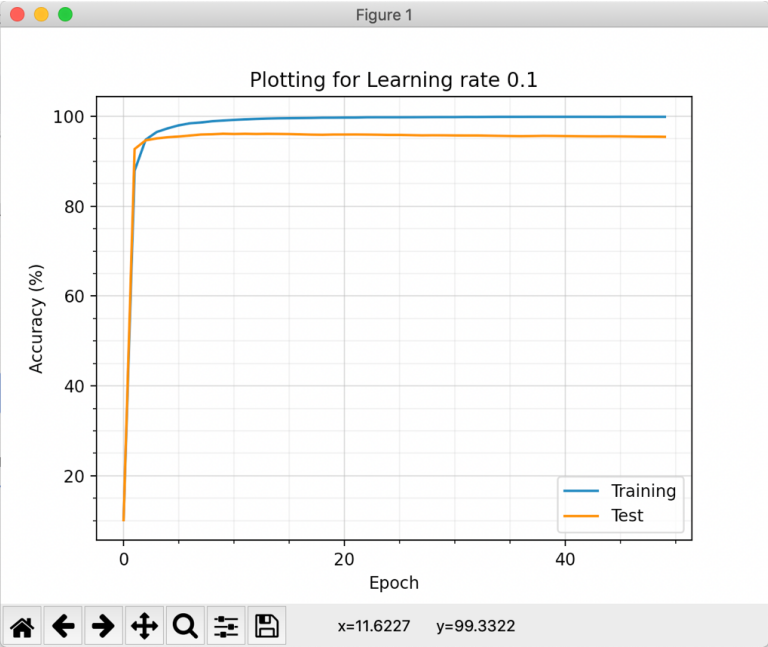
Experiment-3

- 1. How does the size of the training data affect the final accuracy on the test data?
- 2. How does it affect the number of epochs needed for training to converge?
- 3. Again, is there evidence that any of your networks has overfit to the training data? If so, what is that evidence?

One quarter of dataset:



One half of dataset:



Observation:

Accuracy gets closer to 100 when size of the data expands. Basically, more accurate prediction with half the size of the data than the quarter part of the data. When the size of the dataset was half, less number of epochs were required to train than the case where dataset size was 1/4th. With one half of training examples the training data converges after 10 epochs. Networks have overfit to the training data as there is no much fluctuations between training accuracy and testing accuracy.